

Review

AI-enhanced digital twins in maintenance: Systematic review, industrial challenges, and bridging research–practice gaps

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ABSTRACT

The convergence of artificial intelligence (AI) and digital twin technology is reshaping maintenance strategies in the era of Industry 4.0. However, gaps persist between academic advancements and industrial adoption and expectation. This study systematically investigates the landscape of AI-enhanced digital twins for maintenance by integrating a systematic literature review (SLR) of related studies with in-depth interviews from industry practitioners. Our analysis reveals that while academia demonstrates robust applications of supervised, deep, and reinforcement learning to optimize digital twin models and prescribe data-driven actions, industrial implementation remains limited by challenges such as high scale dimension, data integration complexities, and insufficient workforce readiness. We identified and articulated three critical gap dimensions, scale, data, and model between academic research and industrial implementation and expectation. To bridge these gaps, we proposed a comprehensive five-layer framework for AI-enhanced digital twins, encompassing physical assets, data transmission, digital twins, AI analytics, and maintenance services. Actionable recommendations are provided, including the adoption of modular architectures, standardized data protocols, hybrid edge-cloud solutions, and targeted workforce upskilling. Our findings not only clarify the current state and challenges of AI-driven digital twins in maintenance but also offer a practical roadmap for accelerating their industrial implementation. This work advances the field by integrating insights from both academic research and industrial practice, offering concrete recommendations to support the practical realization of smart and sustainable maintenance practices.

1. Introduction

The efficiency and reliability of industrial systems are critical for maintaining productivity and ensuring operational safety. Despite advancements in maintenance strategies, unplanned downtime and equipment failures remain common, leading to significant financial losses and inefficiencies [1]. For example, the total annual costs associated with maintenance were estimated at \$222 billion in 2021 using Monte Carlo analysis [2], while overall equipment effectiveness (OEE) has stagnated at around 50% [3]. These persistent challenges highlight the need for more effective maintenance approaches.

In response, digital twins have emerged as a cornerstone of Industry 4.0, enabling real-time monitoring and predictive analytics for maintenance practices [4]. By creating dynamic, real-time virtual models of physical assets, digital twins enable better decision-making through simulation and analysis of sensor data and operational systems [5]. Their integration with artificial intelligence (AI) has further expanded capabilities, shifting digital twins from passive simulation tools to

autonomous systems capable of intelligent decision-making in maintenance [6–10].

However, while AI-enhanced digital twins show great promise for advancing maintenance practices, their application in practical industrial settings is still in its early stages [5], and their adoption in real-world industrial settings remains constrained by the absence of practical guidelines and tools to support implementation [11]. This gap between academia and industry suggests challenges in translating theoretical advancements into tools that industries can readily implement [12,13].

This study aims to address two key objectives. First, it seeks to explore how AI can augment digital twins to support maintenance tasks and address academic knowledge gaps. Second, it aims to perform a comprehensive gap analysis of AI-enhanced digital twins by examining differences between academic research and industrial practices while identifying barriers that hinder their real-world implementation. To achieve these objectives, this study will combine a systematic literature

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review (SLR) with interviews with industry practitioners to analyze these gaps. Specifically, it will address the following research questions:

- What types of AI have been applied to digital twins to support maintenance in industry, and how are they utilized for enhancing digital twins in academic research?
- What are the gaps in terms of applications, technologies and expectations for AI-enhanced digital twins in maintenance between academic research and industrial practices?

By identifying these gaps and understanding the barriers to adoption, this study will accelerate the transition of AI-enhanced digital twin technologies from theoretical concepts to practical applications. Addressing these challenges will help industry stakeholders adopt innovative solutions more seamlessly and move closer to achieving failure-free operations and near-zero maintenance paradigms.

Our study distinguishes itself from existing literature reviews in several critical dimensions. While previous reviews such as [5,14–17] have examined digital twins and AI applications broadly, our research specifically focuses on gap analysis between academic research and industrial implementations and expectations within the manufacturing maintenance domain. Our integrated methodology uniquely combines systematic literature analysis with empirical interview data, enabling identification of specific AI and digital twin platforms suitable for industrial implementation while providing comprehensive understanding of both theoretical advances and practical adoption barriers that existing reviews have not addressed. Specifically in literature review, while studies [5,16,17] focus on digital twins supporting predictive maintenance, our study emphasizes and analyzes AI that enhances digital twins for maintenance applications. Study [15] concentrates on predictive maintenance in the mining field, whereas our research provides broader coverage of manufacturing maintenance contexts. Furthermore, while study [14] discusses the role and function of machine learning for digital twins, our study additionally examines digital twin platforms optimized for AI scalability and their industrial implementation challenges. These distinctive features position our work as a comprehensive bridge between academic advances and industrial realities in AI-enhanced digital twin applications.

The paper is organized as follows: Section 1 introduces research background and research questions. Section 2 explains the theoretical background of this study. Section 3 outlines the research methodology, including a SLR for academic perspective and interview design for industrial perspective. Section 4 presents findings from the SLR, which provide a foundation for understanding the current state of the art, along with interview results that reveal the industrial applications' current status, challenges, and outlook. This section then identifies key gap dimensions. Section 5 discusses answers to research questions, proposes a theoretical framework based on SLR results, and presents actionable recommendations. Finally, Section 6 summarizes the study's contributions and conclusions.

2. Theoretical background

2.1. Maintenance development and smart maintenance

Maintenance practices have evolved significantly over the past decades, progressing from reactive approaches to advanced strategies such as preventive, condition-based, predictive, and prescriptive maintenance [5,18]. Reactive maintenance addresses equipment failures only after they occur, often leading to unplanned downtime, operational disruptions, and high repair costs [19]. To mitigate these drawbacks, preventive maintenance was introduced, relying on scheduled inspections and component replacements based on fixed time intervals or usage metrics, thereby reducing the frequency of unexpected failures [20]. Further advancements led to condition-based

maintenance, which monitors equipment health in real time and performs maintenance only when necessary, optimizing resource allocation and minimizing unnecessary interventions [21]. The integration of advanced sensor technologies and computational analytics has enabled predictive maintenance, which leverages machine learning to forecast potential failures and facilitate proactive interventions [1]. Most recently, prescriptive maintenance has emerged as the next evolutionary step, integrating operational data, contextual constraints, and optimization algorithms not only to predict failures but also to prescribe actionable, data-driven failure mitigation strategies [22,23]. By adopting these advanced methodologies, maintenance practitioners can proactively identify and address potential issues, thereby reducing downtime and enhancing operational efficiency.

The concept of smart maintenance encompasses various interpretations across the literature. Some studies adopt a technical perspective, framing it as the integration of advanced digital technologies and data-driven methodologies into conventional maintenance systems to enable predictive and prescriptive maintenance capabilities [24,25]. Others conceptualize it more broadly as an organizational framework for managing maintenance in digitally transformed manufacturing environments, emphasizing the synergistic interplay of data, human expertise, and cross-functional collaboration to improve decision-making [26,27]. This study aligns with the technical interpretation, focusing on how smart maintenance leverages real-time monitoring, automated fault diagnostics, predictive analytics, and remote control functionalities to enhance equipment reliability and operational productivity [28]. By adopting such technology-centric approaches, organizations can optimize maintenance efficiency while significantly reducing unplanned downtime.

2.2. Digital twin

A digital twin is a dynamic, data-driven virtual representation of a physical entity, system, or process, enabled by real-time synchronization and computational analysis [29]. It integrates multi-domain models, sensor data, and machine learning to emulate, predict, and optimize the performance of its physical counterpart [30]. The core components of a digital twin include a physical asset with embedded sensors, a virtual model with physics-based or data-driven simulations, and bidirectional data transmission that connects the physical and virtual models [31].

Digital twin design patterns address specific life cycle stages in system engineering [32]. A digital model operates offline without real-time data, while a digital shadow reflects unidirectional updates from the physical system to the virtual system. A digital twin achieves bidirectional synchronization, enabling the virtual entity to influence the physical system through closed-loop control [33]. Additionally, digital twins are hierarchically classified by scope: equipment level (or component level) represents a single machine or its components, and system level represents an assembly line or an entire factory (some studies refer to these as system-of-system level or factory level, but this paper does not distinguish them) [5]. These classifications highlight the scalability of digital twins, which can be modularly adapted for applications ranging from real-time monitoring to what-if scenario testing [34].

2.3. Artificial intelligence

AI has gained significant traction in industrial applications due to its capacity to simulate cognitive functions through algorithmic and data-driven models, enabling autonomous operations. A key characteristic of AI systems lies in their adaptive capabilities that they can process new inputs, extract generalized knowledge from training data, and continuously enhance their performance through iterative learning processes [35]. The field of AI encompasses various technical approaches classified by their learning methodologies. Machine learning represents

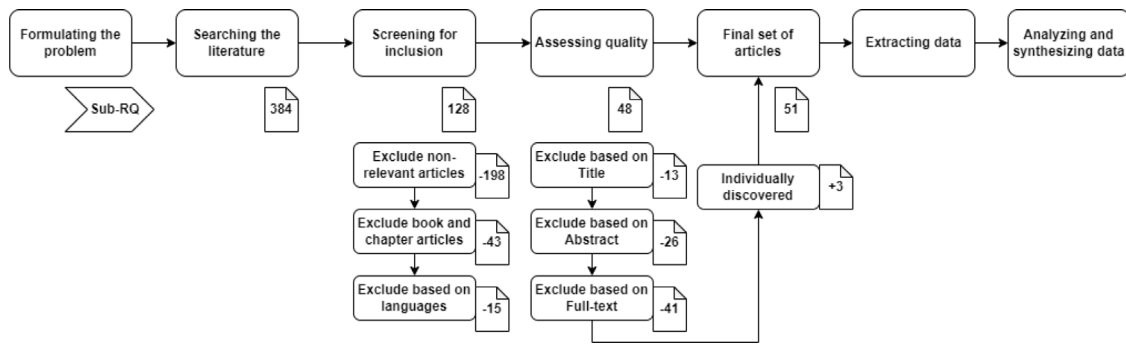


Fig. 1. Literature search process.

the predominant approach, consisting of three fundamental paradigms. Supervised learning employs labeled datasets for tasks such as regression and classification [36]. Unsupervised learning discovers inherent patterns in unlabeled data through techniques like clustering [37]. Reinforcement learning enables systems to optimize actions based on environmental feedback mechanisms [38]. More sophisticated machine learning implementations, particularly deep learning architectures using neural networks, demonstrate exceptional proficiency in handling complex data structures including image data and temporal sensor measurements [39]. This capability renders them especially valuable for predictive maintenance systems and digital twin implementations.

Modern AI implementations use several critical supporting technologies to achieve scalability. Cloud computing platforms offer flexible computational resources for large-scale model training, while distributed systems frameworks facilitate parallel processing of extensive datasets [40]. Edge computing architectures provide localized processing capabilities for time-sensitive inference tasks [41]. These technological components interact synergistically with AI algorithms to enhance digital twin functionalities and optimize maintenance decision-making processes.

3. Research methodology

This study integrates a SLR and interviews with industry practitioners to identify gaps between academic research and industrial practices in AI-enhanced digital twin applications for maintenance. The SLR is chosen for its rigorous and structured process, ensuring a comprehensive analysis of existing academic research [42], offering a structured examination of theoretical advancements, while the interviews are included to capture firsthand perspectives from professionals actively engaged in implementing these technologies. This approach provides a direct comparison between academic trends and real-world challenges, ensuring a balanced perspective.

3.1. Systematic literature review

A review protocol was developed before conducting the SLR in alignment with guidelines [42,43]. Fig. 1 illustrates the literature review process we adopted.

3.1.1. Formulating the problem

We structured our literature search to find essential research about AI-enhanced digital twins in maintenance systems. To maintain focus, we intentionally excluded studies that do not directly address this combination.

Studies were selected for inclusion based on their explicit integration of maintenance strategies, such as reactive, preventive, condition-based, predictive, or prescriptive approaches, with digital twin frameworks. To ensure clarity, works were required to explicitly reference digital twin terminology (e.g., “digital twin”, “virtual twin”) and incorporate AI-driven methodologies, including machine learning, deep

learning, neural networks, or related techniques. Given the rapid evolution of AI and digital twin technologies, priority was given to publications from January 2019 to August 2024 to capture state-of-the-art advancements since SLR is common in AI field to choose 5 years data. Non-English publications were excluded to mitigate resource constraints associated with translation and ensure consistency in analysis.

Articles were searched using the Scopus, using an “all-inclusive” query. All inclusive query is shown below:

(TITLE-ABS-KEY (“artificial intelligence” OR “AI” OR “machine learning” OR “deep learning” OR “neural networks” OR “intelligent systems” or “reinforcement learning”) AND TITLE-ABS-KEY (“digital twin” OR “digital twins” OR “digital model” OR “digital models” OR “digital shadow” OR “virtual twin” OR “virtual twins” OR “virtual model” OR “virtual models”) AND TITLE-ABS-KEY (“predictive maintenance” OR “prescriptive maintenance” OR “reactive maintenance” OR “proactive maintenance” OR “preventive maintenance” OR “Condition based maintenance” OR “Smart Maintenance”))

The query encompasses three key clusters of terms: AI, digital twins, and maintenance. This approach ensures alignment with the study’s focus on the intersection of these domains. The inclusion and exclusion criteria outlined above were applied to filter articles for quality assessment, excluding non-relevant articles, books, book chapters, and publications in languages other than English. The paper should present an actual implementation instead of only an idea. Some studies have English title and abstract, while the full content is in non-English, for example study [44]. The scope of maintenance in this study is specifically limited to the maintenance service of machinery and production systems, excluding network system maintenance or other forms of IT infrastructure maintenance. For instance, study [45] was excluded because the maintenance field is about mechanical and electrical plant’s malfunctioning for residential technical plant systems.

During screening process, titles and abstracts were initially screened to determine the relevance of each article to our research scope. While some articles included relevant keywords in their titles or abstracts, a full-text review revealed that many did not meet our criteria. For example, some studies discuss digital twins solely as an emerging technology in a theoretical context without applying them to maintenance practices [46]. Additionally, studies lacking AI methods to support digital twin applications, as well as review papers that discussed concepts without presenting practical implementations, were also excluded. For example, study [47] was removed because it does not discuss how they enhance digital twins with AI models. Furthermore, to ensure comprehensiveness, individually discovered papers that were initially missed during the screening process were also included in the final set of articles, for example, study [48] was individually found through in Google Scholar due to its high relevance to the research topic. It was missing from the first screening, because the abstract does not mention about the term “maintenance”, while the topic “anomaly detection” is an important topic in maintenance.

3.1.2. Extracting data

The data extraction process involved a thorough review of the full text of each selected article to identify and systematically tabulate key information relevant to our research objectives. The extracted data encompassed a wide range of dimensions, including the geographic distribution of research activity by country, and the specific keywords and fields addressed within the literature. Industry fields analyses were conducted to understand the application contexts of digital twins, while the types of maintenance strategies discussed were categorized to align with our research scope. Furthermore, we documented the physical systems associated with digital twins, the types and maturity levels of digital twin models, and the software, tools, and protocols utilized by authors to develop these models. Special attention was given to the AI tools and algorithms integrated into digital twin frameworks, as well as the maintenance objectives addressed in each study. Proposed frameworks and methodologies were also extracted to evaluate their applicability and innovation. Finally, we identified and recorded research gaps highlighted by the authors to provide a comprehensive understanding of the current state of the field and areas requiring further exploration.

3.1.3. Analyzing and synthesizing data

The extracted data were systematically analyzed to address the five objectives of this academic systematic review. First, the analysis identified industries and application domains adopting AI-enhanced digital twins for maintenance, with a focus on the different emphases of maintenance objectives. Second, the platforms and tools employed for scalable digital twin modeling were evaluated to identify trends in simulation and digital twins method, including protocols and simulation software that support seamless AI integration. Third, the AI techniques applied within digital twin systems were categorized, distinguishing between general machine learning algorithms, such as supervised learning and unsupervised learning, and advanced methodologies like deep neural networks. Fourth, the analysis assessed how different AI models enhances digital twins' capabilities to better support maintenance objectives. Finally, different proposed frameworks that integrating AI and digital twins were synthesized to evaluate their structural coherence, scalability, and alignment with industry-specific maintenance requirements. Each selected paper was analyzed according to the five research objectives for further processing.

3.2. Interview study

To collect the insights from industrial side, we conducted several individual interviews with industrial stakeholders that knowledgeable in maintenance domain, and then analyzed interview transcriptions using thematic analysis [49].

3.2.1. Interviewee pool and selection criteria

The interview participants were sampled using pre-defined criteria to ensure diversity in industry representation, professional experience, and direct relevance to maintenance practices and digital twin adoption. Following established interview methodology for qualitative research [50], three criteria were employed for participant selection: (1) Participants should hold positions that involve strategic or operational decision-making regarding digital twin adoption, AI implementation, and maintenance technology investments in their organizations. (2) Participants should have comprehensive understanding of both AI technologies and digital twin applications in industrial maintenance contexts, evidenced through their professional responsibilities or project involvement. (3) Participants should have hands-on experience with digital twin technologies, AI applications in maintenance, or related industrial digitalization projects within their current or previous roles. These selection criteria align with similar interview-based studies on digital twin implementation in industrial contexts [51], ensuring methodological consistency.

Using the three criteria, we identified qualified candidates in Sweden, from which we sampled nine industry practitioners from several leading Swedish companies who best represented our target demographic and research objectives. These nine participants participated in semi-structured interviews, offering extensive and multidisciplinary expertise that comprehensively covered the critical themes relevant to AI-enhanced digital twin implementation in industrial maintenance contexts. Detailed participant backgrounds, including qualifications and expertise, are summarized in Table 1. The table also demonstrates participants' alignment with the three selection criteria through their professional roles (criterion 1), technical backgrounds and knowledge (criterion 2), and relevant work experience (criterion 3).

Their roles spanned implementation consultant, CEO, data-driven decision specialist, research manager, AI analyst, maintenance systems engineer, front-line maintenance worker, and simulation team leader, representing application domains such as industrial automation, automotive, software technology, sustainability, consulting and manufacturing systems. Participants held advanced qualifications, including PhDs, and possessed professional experience ranging from four to nearly fifty years. Some of them have both blue-collar and white-collar experience. They contribute deep knowledge in discrete event simulation, simulation-based optimization, data-driven decision-making, asset management and first-hand maintenance experience.

3.2.2. Interview design

The study employed a semi-structured interview approach to capture industry experts' practical experiences, challenges, and forward-looking perspectives on AI-enhanced digital twins in maintenance. This method ensured a balance of structure and flexibility, allowing participants to elaborate on their organizational roles, technical implementations, and strategic priorities while capturing rich, in-depth insights through open-ended questions and follow-up research [52]. Interviews were organized into six thematic phases, each designed to address core dimensions of the research inquiry. The complete list of interview questions is provided in Appendix A.

First, participants' professional backgrounds and industry contexts were established through questions about their roles, managed systems, and specialized use cases. Subsequent phases explored their direct experiences with digital twins in maintenance, including operational examples and evaluations of effectiveness at equipment or system levels. To clarify the role of AI, discussions focused on current AI practices, distinguishing between predictive analytics, decision support frameworks, and integration with digital twin architectures. Participants were then invited to identify technical, organizational, and skill-related barriers to adoption, offering detailed insights into scalability challenges and resource gaps. Finally, interviews explored future expectations and desired advancements and how new tools help maintenance work. Interviews concluded with open-ended reflections to capture unanticipated themes.

3.2.3. Data analysis

During our iterative interview and analysis process, we employed constant comparative analysis [53], continuously comparing new data with emerging findings. By the ninth interview, we observed clear signals of saturation in terms of information repetition across participants, the absence of new major themes emerging from additional interviews, and comprehensive coverage of critical themes and challenges relevant to our research objectives. Owing to these signals, we determined that nine interviews were sufficient to address our specific research questions about AI-enhanced digital twin implementation barriers in maintenance.

Thematic analysis was employed to systematically identify and interpret patterns within the qualitative interview data [49]. This approach involved a six-phase process: (1) familiarization with the data, (2) generating initial codes, (3) searching for themes, (4) reviewing

Table 1
Overview of interviewees’ background, roles, and expertise.

Code names	Role	Background	Work experience
R1 ^a	Implementation consultant	Focus on team center integration, migration, and simulation; also teaches discrete event simulation	6 yr
R2	CEO	Simulation-based optimization, analytics, and AI transformation; PhD and research experience	20 yr
R3	Data-driven decision specialist	Expertise in smart maintenance, data-driven decision-making, and asset management; extensive experience in maintenance operations	47 yr
R4	CEO	Leads an industrial think tank focusing on sustainable production, maintenance, and new technologies; facilitates knowledge sharing among various industries	30 yr
R5	Research manager	Coordinates research in production and logistics; PhD in maintenance management; expertise in data-driven maintenance and electrification challenges	7 yr
R6	Maintenance systems engineer	Deeply involved in automotive maintenance department; expertise in data-driven decisions, IoT, and smart maintenance	25 yr
R7	AI analyst	PhD in system-level maintenance; specializes in AI applications for supply chain, logistics, and manufacturing optimization	4 yr
R8	Maintenance specialist	Started from the front-line maintenance department, experience both as blue-collar maintenance worker, technician, and white-collar analyst and engineer	25 yr
R9	Simulation team leader	Industrial engineering background; expertise in logistics, simulation modeling, and digital twins for various industries	8 yr

^a R = Respondent.

themes, (5) defining and naming themes, and (6) producing the final analysis [49].

First, interview transcriptions were reviewed multiple times. During this familiarization phase, annotations were made to highlight critical insights, contextual differences, and recurring ideas. The data were then categorized as: participant expertise and experience, current applications of AI-enhanced digital twins in maintenance, technical and organizational challenges, critical improvements needed, and future outlook for the technology.

Then codes were combined into broader themes through iterative refinement. For example, challenges such as “organizational barriers” and “skill gaps in deploying AI-digital twins” emerged as sub-themes under the overall category of adoption barriers. Similarly, participant reflections on generative AI’s potential for digital twins and maintenance were grouped under technological trends. Findings integrated key themes into a cohesive framework, connecting implementation challenges with strategic priorities.

4. Results

4.1. Literature findings

The 51 selected research papers are presented in [Appendix B, Table B.4](#), detailing the title, publication year, industry, physical system, AI techniques that support digital twin, and corresponding maintenance issues. In the following Section 4.1.1, we provide an overall analysis of these studies’ characteristics and trends. Section 4.1.2 examines the diverse application domains and research objectives across papers. Section 4.1.3 discusses the common platforms implemented in these digital twin applications. Finally, Section 4.1.4 analyzes the specific AI models employed to enhance digital twins and explores the resulting capabilities these enhancements deliver.

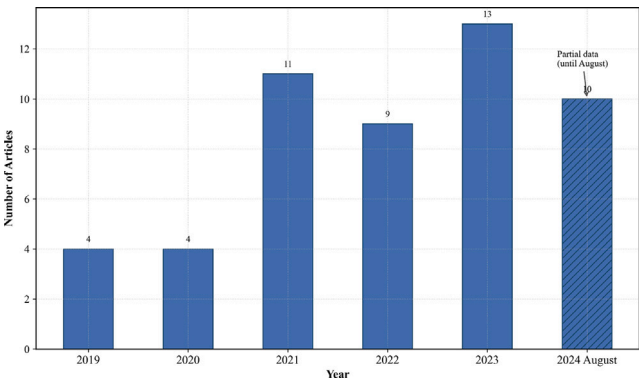


Fig. 2. Number of selected articles by year from 2019 to 2024 August.

4.1.1. Descriptive analysis

Recent studies on AI-enhanced digital twins for maintenance have gained increasing attention, as shown in [Fig. 2](#). The growing number of publications highlights the increasing interest in this field, with more researchers focusing on developing smart maintenance solutions using digital twins. However, since our SLR process began in August 2024, the publication count for 2024 does not fully represent the year’s total output.

[Fig. 3](#) illustrates the global distribution of research contributions, with studies spanning 20 countries across five continents. This geographical diversity underscores the growing relevance of AI-enhanced digital twins for predictive maintenance, as researchers from diverse regions actively advance the field. Notably, China and Italy stand out as leading contributors, each publishing eight significant studies in recent years. While Italy’s contributions focus on frameworks such

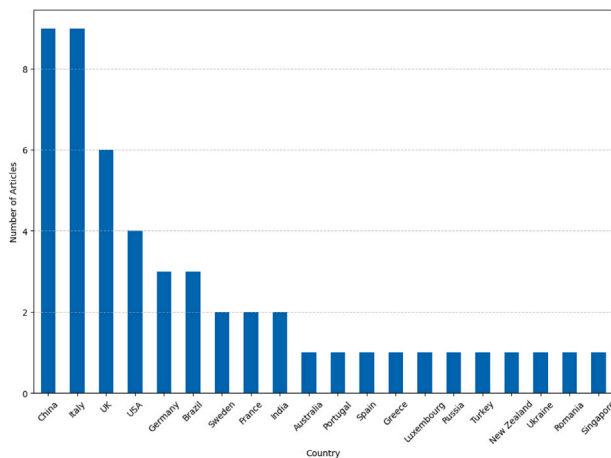


Fig. 3. Number of articles by country for selected studies.

as the IoT Twins [54,55] and REPLICA architectures [56,57], China's publications demonstrate broader thematic diversity. These span maintenance applications in manufacturing [58], railway systems [59], mining [60], aerospace [61], and power grids [62], etc. Such broad topics highlight China's expansive exploration of AI-enhanced digital twins across industries, reflecting the technology's adaptability to diverse maintenance challenges. These efforts collectively emphasize the field's multidisciplinary appeal and global momentum, driven by collaborative innovation from institutions worldwide.

Fig. 4 shows a strong thematic structure. At its center, three keywords form a tightly connected triangular cluster, highlighted by their large node sizes which indicates high frequency and thick connecting lines that shows strong conceptual relationships. Notably, digital twin stands out as the most prominent node, reflecting its central role in the network. Surrounding this core cluster are secondary themes like Industry 4.0, anomaly detection, fault diagnosis, and artificial intelligence, all closely linked to digital twin. This pattern shows its critical role in connecting technical and operational domains.

Over the past five years, 44 out of the 51 cases of AI-enhanced digital twin applications in maintenance have focused on predictive maintenance, with only 2 cases addressing preventive maintenance [63, 64] and 5 case on prescriptive maintenance [48,61,65–67]. This distribution suggests that we are still at an important stage in the transition from predictive to prescriptive maintenance. Thus, when discussing smart maintenance, the emphasis remains predominantly on predictive maintenance.

Regarding the abstraction level, 27 out of the 51 studies apply digital twin at the equipment level, such as milling machine [58], axis test rig [68], hoist motor [60], etc. The remaining 24 studies focus on the entire production systems, where digital twin is utilized at the system level, such as pipelines [69], factories [63,70,71], and production lines [65,72–75].

4.1.2. Application domain & objectives

Fig. 5 illustrates the broad range of real-world contexts in which AI-enhanced digital twins are deployed for smart maintenance, spanning twelve application domains. The most widely studied domain is industrial IoT & machinery, with 12 studies [48,58,68,76–84]. Within this domain, research mainly focuses on manufacturing applications that require real-time data processing and continuous monitoring of machinery health, such as tool wear detection in machining operations [48,58,84]. These studies emphasize integrating sensor-driven digital twins with advanced AI models to enable proactive maintenance and swift adaptation to operational changes. The steel industry is analyzed separately from general machinery and smart factories, with

studies specifically targeting maintenance strategies for production lines [13,74,75] and steel industrial components [85].

The smart factory [54–57,63,65,70,86,87] and energy [62,66,69,71, 88–92] domains follow closely, each with 9 studies. Among smart factory studies, several focus on production lines [56,57,65], while others explore more specialized applications, such as packaging systems [86] and integrated production planning [55,65]. Despite these variations, all studies highlight the role of AI-driven digital twins in leveraging real-time data to streamline maintenance and enhance operational efficiency.

In energy-focused research, one-third of the studies are about oil and gas maintenance [66,69,89], particularly for sub-sea pipelines [66, 69] and drilling equipment [66]. The remaining studies address electric power maintenance, covering transformer fault detection [62,88] and wind turbine upkeep [92]. Across both energy sub sectors, researchers emphasize the importance of real-time, AI-driven prognostics for minimizing downtime, enhancing safety, and optimizing operational costs.

Although the automotive sector features only three studies, they span multiple components, including tires [93], engines [94], and car-body structures [72]. Similarly, in railway maintenance, research covers diesel bearings [59,95] and railway track monitoring [96]. Notably, studies on engine maintenance appear across different sectors, including energy, aviation, and automotive [61,71,94,97], emphasizing the cross-industry significance of AI-driven digital twins in ensuring reliability and efficiency.

The heatmap in Fig. 6 illustrates the application domains and maintenance objectives. On the vertical axis, five key objectives are shown: anomaly detection, fault diagnosis, remaining useful life (RUL) prediction, health index, and the proposed framework, each reflecting recurring strategies in these studies. The categorization of maintenance objectives aligns with prior research [5,14,98].

Anomaly detection is a key maintenance objective that focuses on identifying early irregularities in equipment performance, enabling timely notifications to practitioners for proactive maintenance actions [99]. There are 13 out of 51 studies contains anomaly detection [13,48,59,75,76,87,88,91,92,95,100–102], including time series and image anomaly detection. Fault diagnosis involves analyzing observations, such as sensor signals and mathematical models of a technical system, to detect and identify abnormal system behavior. Its primary goal is to determine the presence of a fault and, if detected, pinpoint its location [103]. Paper [56,59,60,62,63,66,71,72,77,78,80–82, 86,89,97,104] contain fault diagnosis in their maintenance studies. The RUL is a critical indicator for predicting a component's failure before it occurs. RUL prediction can be performed using historical data or direct data extraction, employing model-based, data-driven, or hybrid approaches [105]. 12 studies involves RUL prediction and estimation [58,61,68,69,79,82,83,85,89,93,95,106]. Health index and condition analysis of production equipment are an important step in implementing predictive maintenance, providing company managers with performance and degradation metrics that help predict the condition of components [107]. Health index represents a real-time numerical assessment of a machine's current condition, whereas RUL prediction estimates the time left before the machine reaches failure. There are four studies contains machine's health index method [83, 84,94,97] to monitor machines' in-time state. There are 19 studies that propose a framework to use AI to enhance digital twins to support maintenance [55,55,57,57,61,64,65,67,68,70,73,74,80,81,84,88, 92,94,96].

The heatmap clearly illustrates that anomaly detection is a universal objective across most of application domains. This widespread adoption highlights that industries, regardless of their unique operational challenges, value the ability to quickly identify deviations from normal behavior as a foundation for effective maintenance. In particular, sectors such as energy, industrial IOT & machinery, and smart factory show a pronounced focus on fault diagnosis; these industries

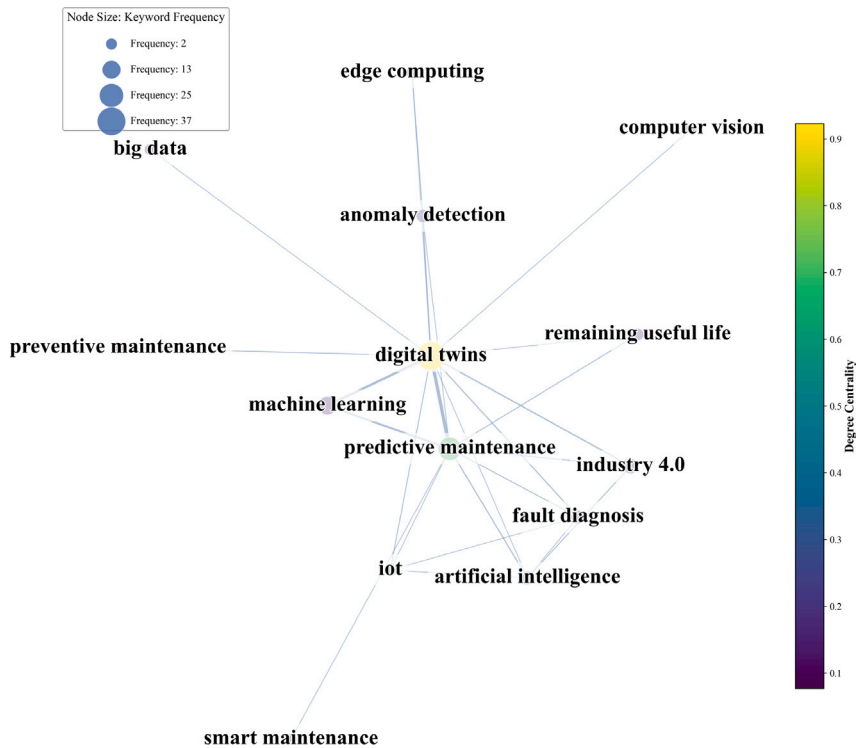


Fig. 4. SLR keyword co-occurrence network.

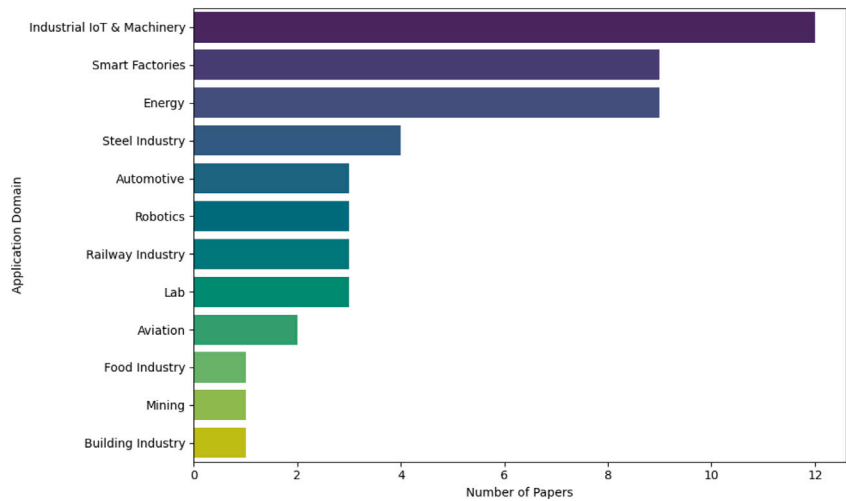


Fig. 5. Application domain of AI-enhanced digital twins for maintenance.

leverage digital twin technology to accurately show faults that could significantly disrupt operations or compromise safety.

Interestingly, smart factory research is marked by a strong inclination toward the development of comprehensive digital twin frameworks, indicating a trend towards integrating predictive analytics, real-time monitoring, and resource management into a unified system. For example, REPLICA framework hosts various AI algorithms in a dedicated AI environment are used to support fault diagnosis and detection [56,57]. Moreover, RUL prediction and health index monitoring are predominantly concentrated in the machinery domain, driven by the availability of detailed, equipment-level digital twin models that enable granular analysis of component wear and performance. This focus on prognostics in machinery not only supports proactive maintenance strategies but also facilitates optimal scheduling, reduced downtime, and improved overall reliability in complex industrial systems.

4.1.3. Digital twins platform for scalability of AI

In selected studies, researchers have utilized a range of scalable digital twin platforms to model, analyze, and optimize complex systems. These platforms integrate cloud computing, AI, internet of things (IoT), and big data analytics, providing robust environments for real-time decision-making and predictive maintenance. These different digital twin platforms can be categorized into four broad categories: commercial/industrial digital twin platforms; general modeling tools; IOT/cloud infrastructures; in-house or custom solutions.

ThingWorx by PTC is a key industrial IOT platform that enables organizations to connect, build, analyze, and manage digital representations of physical assets, providing real-time visibility and data-driven insights essential for creating and maintaining effective digital twins [108]. Study [79] utilized Thingworx Kepware Server [109] for data collection, ThingWorx for IoT integration, and Grafana [110] for

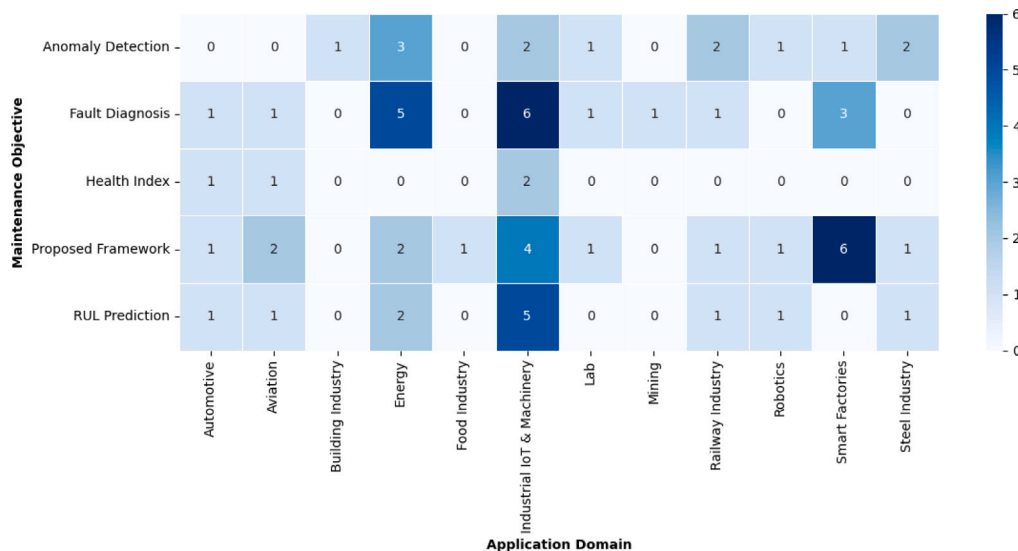


Fig. 6. Application domain vs. maintenance objectives.

visualization, while Grafana was also used in study [55]. Similarly, Siemens platforms play a crucial role in digital twin development, offering strong scalability for AI applications. Study [64] employed Siemens Process Simulate [111] to create robotic digital twins, whereas [67] developed a drone factory digital twin using Siemens Plant Simulation [112]. Beyond these platforms, AnyLogic [113] is widely adopted for production line digital twins, as seen in studies [65,82,86]. Meanwhile, its competitor, FlexSim [114], has also been applied in industrial digital twin applications, with studies [70,97] leveraging it for system-level learning in industry and equipment-level hydraulic accumulator digital twins.

As versatile modeling tools, Matlab [115] and Simulink [116] are widely recognized for their programming capabilities and are utilized as powerful platforms for developing and implementing digital twins [77,80,89,101,106]. They excel in creating physics-based and control-system models, with seamless integration of signal processing and machine learning toolboxes. Unity [117] is emerging as a leading platform for building immersive and interactive digital twin simulations across industries. Unity's strength lies in its ability to connect with AI tools, offering easy integration with deep learning frameworks like TensorFlow and PyTorch, enabling developers to create intelligent and adaptive digital twins [118]. For instance, Liang et al. [60] leveraged Unity to develop a high-fidelity digital twin monitoring system for mine hoists, enabling real-time temperature data acquisition and integrating neural networks for feature extraction and precise motor state prediction.

Open Platform Communications Unified Architecture (OPC UA) and Message Queuing Telemetry Transport (MQTT) are two widely used communication protocols in industrial automation and IOT field. OPC UA is a platform-independent, service-oriented architecture protocol that enables secure and reliable exchange of process data, alarms, events, and historical information between clients and servers [119]. Studies [81,104] applied OPC UA to capture the data from the IoT sensors to create digital twins. MQTT is a lightweight, publish-subscribe messaging protocol designed for machine-to-machine communication in low-bandwidth environments, allowing devices to publish messages to specific topics and subscribe to receive information through a central broker [120]. Studies [76,92] used this lightweight messaging protocol MQTT to create digital twins.

IOTwins [54,55] and REPLICA [56,57] are two distributed and scalable architecture for intelligent digital twin realization, combining edge/cloud computing and integrated AI workflows for fault diagnosis and predictive maintenance. There are also some custom or private

infrastructure as a service (IaaS), employing deep neural networks and classification/clustering for resource allocation and anomaly detection [100]. Node-RED [121] is a flow-based tool built on Node.js and it is often used for custom, cloud-based digital twin data pipelines [97]. Additionally, finite element models are also used to build as digital twins. Bhowmik [69] built their own finite element models for machine learning based analytics to make informed maintenance decisions, ensuring improved pipeline reliability and safety.

4.1.4. AI category & capability

Table 2 shows the algorithms that selected studies have used and the corresponding categories. There are overall 7 different AI categories represented in the selected studies, ranging from general deep learning, transfer learning, reinforcement learning, and clearly defined supervised, unsupervised, and semi-supervised learning approaches, optimization techniques, statistical models, supporting technologies, and general/hybrid machine learning methods as we used the AI classification method from paper [122]. Our categorization aligns with prior taxonomies [123,124] but prioritizes the primary methodological focus of each study. Consequently, overlaps may arise—for instance, some deep learning-based algorithms are classified under supervised learning when their core contribution emphasizes traditional supervised paradigms.

• Deep learning

In the deep learning category, recurrent neural networks (RNN) and long short-term memory (LSTM) are the most widely used algorithms applied to digital twins for maintenance in the selected studies. RNN is a neural network designed to process sequential data, while LSTM, as a variant of RNN, employs gating mechanisms to overcome the vanishing gradient problem during long-sequence training, thereby better capturing long-term dependencies [125]. Recent studies have demonstrated the effectiveness of RNN and LSTM in predictive maintenance. For instance, studies [68,69,79] utilized these algorithms to predict the RUL of equipment, while Liang et al. [60] applied LSTM to forecast temperature variations. Additionally, Abdullahi et al. [92] leveraged LSTM to provide predictive analytics and insights into the performance of wind turbine components. By integrating real-world data into digital twin models, RNN and LSTM significantly enhance the predictive capabilities of digital twins, enabling more accurate and proactive maintenance strategies. As for other sequential models, temporal convolutional

Table 2
AI algorithms and categories that enhance digital twins in selected maintenance studies.

AI category	Sub category	Specific AI algorithms (Key references)
Deep Learning	Sequential Models	LSTM [60,68,69,92], RNN [79], TCN [79], Stacked/Bi-Directional GRU [58]
	Spatial Models	CNN [66,79,104]
	Attention-Based Models	Transformer (TFT [93], DPformer [61])
	Transfer Learning	Deep Transfer Learning [72,77], Few-Shot Learning [104]
Supervised Learning	Regression Models	Linear Regression [85], Poisson Regression [85], Neural Network Regression [85], Decision Tree Regression [85], Boosted Decision Tree Regression [85]
	Classification Models	Logistic Regression [70], SVM [68,70,78,104,106], KNN [78], Decision Tree [78,80], Kernel Extreme Learning Machine (KELM) [62], Naive Bayes [78,97]
	Ensemble & Boosting Methods	Random Forest [78,97], XGBoost [74,75,92], Gradient Boosting [71], LightGBM [97], Boosted Decision Tree Regression [85]
	Neural Network-Based Models	ANN (shallow nets) [70,71], Nonlinear AutoRegressive with eXogenous (NARX) [48,101], Siamese Neural Network [104]
Unsupervised & Semi-Supervised Learning	Clustering	Clustering [84]
	Feature Learning	Autoencoder [68], Siamese Autoencoder [87], Novel Sparse De-noising Autoencoder (NSDAE) [77]
	Generative Models	Semi-Supervised GAN [102]
Reinforcement Learning	–	General RL [73]
	Deep Reinforcement Learning	PPO [67], A2C [96]
Statistical Models	–	Statistical Models [82,86], Regression Analysis [13]
Supporting Technologies	Distributed/Edge Technologies	Cloud Computing [54,59,76,78,88,100], Edge Computing [55,59,68,73,76,80], Distributed AI (decentralized) [54]
	–	Acumos AI Modules [56], MATLAB Toolbox [80]
General/Hybrid ML	–	[57,64,73,89]

network (TCN) addresses the limitations of traditional RNNs by using dilated convolutional to capture long-term dependencies in sequential data [126], and it is applied in study [79] for monitoring machine’s health and enhance the digital twin’s predictive capability as well.

Convolutional Neural Networks (CNNs) are spatial models primarily designed for processing data with spatial structures, such as images and videos. Their core strength lies in leveraging convolutional operations to extract local features and capture global information through hierarchical structures [127]. For instance, Bhowmik [66] utilized CNNs to analyze in-line inspection images for corrosion detection and classification, while Kumar et al. [104] applied CNNs to classify faults using vibration signals converted into 2D images. By enabling real-time, automated analysis of multi-modal data (e.g., images and sensor inputs), CNNs enhance digital twin capabilities, providing accurate defect detection and predictive maintenance.

Transformer is an attention-based model that captures dependencies in sequences through self-attention and multi-head attention mechanisms [128]. These mechanisms enable the model to dynamically focus on the most relevant parts of the input data,

making it highly effective for tasks involving sequential or structured data. For instance, Mao et al. [61] applied a dual-sense pyramid transformer to analyze both spatial and temporal features from sensor data for predicting the RUL of equipment. By leveraging attention mechanisms, the model captures complex dependencies within the data, significantly improving the accuracy of predictions. Similarly, Karkaria et al. [93] utilized attention mechanisms combined with variable selection networks to prioritize the most informative input features, enhancing the model’s ability to make dynamic maintenance decisions. The integration of transformers into digital twins enables high-fidelity simulations and real-time decision-making, further improving maintenance efficiency and system reliability.

Transfer learning is a machine learning approach that transfers learned knowledge from one task or domain to another related task or domain. It is particularly useful in scenarios with limited data or restricted computational resources, helping to reduce training costs and enhance model generalization [129]. Xu et al. [72] employed stacked sparse autoencoders combined with deep transfer learning to mitigate data distribution shifts and address limited training data, applying this method to a car

body-side production line for real-time fault detection and life-cycle management. Similarly, Xia et al. [77] integrated a novel sparse de-noising autoencoder (NSDAE) with parameter transfer learning, leveraging simulated data from digital twin models to fine-tune a diagnostic model for a triplex pump, achieving high-accuracy fault detection under varying system conditions. In cases where only minimal data is available, few-shot learning, a specialized form of transfer learning, can still be effective by relying on pre-trained knowledge to improve model generalization [130]. Kumar et al. [104] applied few-shot learning with Siamese neural networks to vibration signal-based fault classification, where vibration data was converted into 2D images. This approach enabled digital twins to predict faults in rotary assets, such as motors and fans, using minimal training samples, significantly reducing the need for extensive labeled datasets.

• Supervised learning

Among the selected studies, the majority of AI algorithms belong to the supervised learning category, including regression models, classification models, ensemble/boosting methods, as well as neural network-based approaches. D'Urso et al. [85] applied different regression algorithms to predict RUL of steel components, including linear regression, Poisson regression, neural network regression, decision tree regression, and boosted decision tree regression. Regression models are statistical techniques used to analyze relationships between variables and make predictions based on data trends [131]. They enhance the capabilities of digital twins by allowing for real-time simulations and the generation of synthetic datasets that can model the aging process of steel components under various loading scenarios. This enables better predictions regarding the RUL, allowing maintenance activities to be scheduled more effectively before a failure occurs. The integration of the dynamic reliability digital twin with regression models helps in understanding the non-linear behavior of materials, leading to more accurate assessments and timely interventions.

Support vector machine (SVM) has been widely integrated into digital twins for predictive maintenance and real-time fault detection in various industrial settings [68,70,78,106]. Leveraging kernel functions (e.g., linear, polynomial, and RBF), SVM effectively handles nonlinear problems by mapping data into higher-dimensional spaces [132]. For instance, Mourtzis et al. [106] identified a medium Gaussian SVM achieving 98.24% classification accuracy for robotic components, while Avornu et al. [78] employed SVM alongside Naive Bayes, K-Nearest Neighbors (KNN), and decision trees to fuse and analyze sensor data. Although SVM often delivers robust results, it can be slow to train and requires considerable parameter tuning. To address these limitations, Yao et al. [62] proposed Kernel Extreme Learning Machine (KELM) that offers faster training and improved fault diagnosis accuracy for oil-immersed transformers. By integrating KELM into a digital twin, the framework supports real-time interaction between the physical transformer and its virtual replica, thereby enhancing both diagnostic accuracy and overall system reliability.

Boosting methods are widely adopted in the selected studies as ensemble learning approaches that combine multiple weak learners sequentially to improve predictive accuracy and reduce bias [133]. For example, Deon et al. [71] employed gradient boosting to classify faults, predict operational deviations, and support decision-making in thermal power plants. However, gradient boosting can be computationally demanding and prone to overfitting, prompting the development of more efficient variants. XGBoost emerged as a scalable implementation that integrates L1/L2 regularization to curb overfitting and parallel processing to accelerate training [134]; Studies [74,75] further enhanced XGBoost with techniques like tree penalization, leaf-node shrinking,

and Newton boosting for anomaly detection in hot-rolling mill production lines, while Abdullahi et al. [92] applied it to model failures of wind turbines. LightGBM, by contrast, significantly lowers computational overhead while preserving high accuracy, making it well-suited for large datasets [135]; Süve et al. [97] demonstrated this by training LightGBM on streaming data to forecast equipment status and failures.

• Unsupervised and semi-supervised learning

Comparing to supervised learning methods, unsupervised and semi-supervised learning studies are much fewer. Clustering is a type of unsupervised learning which can group data points into clusters based on similarity without using labeled data. It automatically discover structure in data and make it useful for exploratory data analysis, pattern recognition, and anomaly detection [136]. Pulikottil et al. [84] employed clustering techniques for data labeling and classification methods for predicting tool wear from a human immune system perspective.

Autoencoders offer a powerful unsupervised learning approach by compressing inputs into a lower-dimensional latent space and then reconstructing them, thus facilitating efficient data representation [137]. For instance, Uhlmann et al. [68] utilized a simple encoder-decoder architecture to support classification in cyber-physical production systems, while Castellani et al. [87] proposed a siamese autoencoder (two autoencoders sharing the same weights) to handle digital twin-generated data alongside minimal real-world labeled anomalies for energy consumption monitoring under weak supervision. Building on the denoising autoencoder, Xia et al. [77] introduced a NSDAE by incorporating sparsity constraints, Swish activation, and maximum correntropy cost to enhance performance in noisy environments. Subsequently, parameter transfer learning was applied to fine-tune the NSDAE with limited target-domain data, using digital twins of a triplex pump to generate fault diagnosis datasets.

Generative adversarial network (GAN) is an adversarial model composed of a generator and a discriminator, trained competitively to produce realistic data [138]. The generator synthesizes fake data to fool the discriminator, while the discriminator aims to distinguish real samples from the generated ones. Hu and Cai [102] proposed a semi-supervised GAN approach for failure prediction, effectively leveraging both labeled and unlabeled data through a generator-classifier structure. This method improves predictive accuracy without requiring extensive labeled datasets, thereby enhancing the dynamic interplay between physical assets and their digital counterparts.

• Reinforcement learning

Few selected studies [67,73,96] have applied reinforcement learning (RL), despite its widespread use in the predictive maintenance field, as shown in [98]. This is because RL, as a trial-and-error-based method, requires an agent to interact with the environment to learn an optimal policy that maximizes cumulative rewards [139]. Discrete event simulation environment can serve as environment of RL. However, most RL environments are simulation-based rather than integrated into digital twins of physical systems.

Eppinger et al. [73] applied RL techniques to train control systems within a simulation environment before live deployment in food production. However, instead of leveraging deep neural networks, this approach relied on dynamic programming, which limits the state space dimensions and reduces adaptability and generalization. In contrast, deep reinforcement learning (DRL) utilizes deep neural networks to approximate value functions, policy functions, or environmental models, enabling it to handle complex tasks in high-dimensional state spaces [140]. Advantage Actor-Critic (A2C) is a DRL algorithm that integrates policy optimization (Actor) with value estimation (Critic), using an advantage function to stabilize learning and improve efficiency [141].

This makes A2C particularly effective in dynamically optimizing maintenance strategies by continuously learning from real-time sensor data, facilitating proactive fault detection and minimizing downtime. Sresakoolchai and Kaewunruen [96] applied A2C to optimize maintenance strategies for railway infrastructure. Meanwhile, Proximal Policy Optimization (PPO) enhances A2C by offering better stability and sampling efficiency [142]. Chen et al. [67] further utilized PPO to prioritize maintenance tasks based on cost models within a digital twin for a lab-scale drone factory, demonstrating its potential for efficient predictive maintenance.

• Statistical models and supporting technologies

Statistical models are integrated to digital twins to process sensor data, extract condition indicators, and estimate RUL to enhance fault detection and optimize maintenance schedules. Harries et al. [82] employed a hybrid statistical model that combines real-time factory data with synthetic data to reduce reliance on historical records, making it more adaptable for small and medium-sized enterprises. Hassan et al. [13] applied regression-based degradation models to predict machine failure trends, while others focus on anomaly detection through statistical deviations between physical machines and their digital twins. Collectively, these studies demonstrate that statistical modeling enhances predictive maintenance by improving failure prediction accuracy, reducing downtime, and supporting proactive decision-making. Some studies do not incorporate specific algorithms to enhance digital twins but instead utilize supporting technologies such as Acumos AI modules [56] and the Matlab Toolbox [80]. Additionally, distributed and edge technologies are widely applied to support digital twins. For instance, cloud computing [54,59,76,78,88,100] efficiently trains and deploys deep learning models by integrating historical and real-time data from sensors and IoT devices, providing distributed storage and unified data management to enhance predictive maintenance. Cloud computing can be further enhanced to support digital twins through advanced optimization frameworks, such as evolutionary multitasking optimization with adaptive knowledge transfer mechanisms that enable efficient scheduling of constrained cloud workflow tasks [143]. Additionally, knowledge transfer-enabled algorithms facilitate diverse task scheduling for individualized requirements in industrial cloud environments, allowing multiple workflow applications to be optimized concurrently while leveraging inter-task knowledge sharing to improve scheduling efficiency and resource utilization [144]. Such enhanced cloud computing approaches can be further integrated with IoT Twins frameworks to support distributed AI solutions and enable predictive maintenance through real-time process and fault predictions in industrial manufacturing environments [54,55].

Meanwhile, edge computing [54,55,59,73,76], as a distributed computing model, brings computational power closer to the data source, significantly reducing latency in data transmission. For example, Raspberry Pi [68,80] is often used to enable real-time responsiveness in digital twin applications. Compared to cloud computing, edge computing excels in low-latency processing, mitigating network delays and localizing computational tasks to ensure critical operations can be executed on-site. Additionally, it reduces network bandwidth consumption, for instance, in smart manufacturing, factory equipment's digital twin can compute health status locally and only upload data to the cloud when abnormalities are detected, minimizing data transmission overhead [54,76]. Furthermore, edge computing enhances system reliability; in factory production lines, it can continue monitoring devices even during network disruptions, preventing shutdowns caused by connectivity failures [55].

4.2. Industrial perspective

The following sections are results for Section 3.2, interview study. It will show consolidated findings and current status of AI-enhanced digital twins in supporting maintenance, the barriers and challenges in industry for applying that, and the expectations from industry and the outlooks.

4.2.1. Consolidated findings and current status

The nine interviewees (R1-R9), based on their own experiential backgrounds, have different views on the manufacturing maintenance as well as some common opinions. Interviewees from automotive original equipment manufacturer (OEMs) (R3, R5, R6, R7, R8) described a decades-long transition from manual, time-based maintenance to data-driven frameworks, often based on centralized platforms for maintenance management, for example, IBM Maximo [145]. Consultants and simulation experts (R1, R2, R9) emphasized the role of discrete-event modeling and AI-driven optimization in bridging gaps between theoretical digital twin concepts and shop-floor realities. For example, one consultant (R2) detailed a prototype combining genetic algorithms with digital twins to prioritize maintenance tasks, while an automotive engineer (R6) highlighted incremental progress toward predictive analytics using real-time programmable logic controller (PLC) data. Despite the different contexts, all participants emphasized the need for AI-enhanced digital twins to move beyond the pilot phase, highlighting their potential for maintenance under a unified IIOT system.

The interviews reveal a gap between the enthusiasm for AI-enhanced digital twins and their real-world adoption. While most participants (R1, R2, R3, R5, R6, R7, R8) recognized the technology's theoretical benefits, such as enabling prescriptive maintenance and real-time simulation, few reported large-scale deployments. As R3 noted, "We've implemented this for larger equipment like robots and PLCs... but it's still a work in progress". However, several industry projects highlighted by respondents demonstrate successful applications. For instance, an automotive engineer described using digital twins to perform minor maintenance during production windows without line stoppages, illustrating how system-level digital twins integrated with AI can address operational challenges (R7). Experimental cases, such as a genetic programming project for maintenance optimization (R2), prove technical feasibility but remain limited to small-scale implementations.

Organizations often settle for "digital shadows" (R3, R4, R7, R9), simplified models that passively aggregate equipment data into dashboards without enabling bidirectional control. As respondents noted, digital twin implementations often reduce to 'dashboarding... related to our KPIs' (R3), where models use historical data inputs to simulate scenarios 'to be rather than something that is' (R3). Similarly, a simulation consultant (R9) observed that clients frequently conflate digital twins with basic simulation models, leading to underpowered solutions that lack AI integration. Participants (R1, R4, R9) noted that industry stakeholders rarely agree on whether a "digital twin" requires real-time data, AI integration, or bidirectional control, further complicating progress.

The development in the field of maintenance demonstrated by the interviewees shows three successive stages of maturity, including data infrastructure, pilot application stage and operational integration of the strategic conceptual framework. First, industries are prioritizing foundational efforts in data infrastructure, deploying IoT-enabled sensors and centralized data platforms (R3, R6), though these systems often function independently of AI models. For example, an automotive OEM (R6) highlighted their cloud-based data repository for aggregating maintenance logs but admitted that predictive analytics still remain manual and reactive. "We have a platform called smartfx, the majority of data runs on Amazon cloud. We have many kind of inputs ... through MQTT ... push to a centralized data platform like data warehouse." Second, pilot projects demonstrate AI's potential in narrow applications, such as anomaly detection (R3) or simulation-based optimization (R2).

R2 cited a prototype where genetic algorithms prioritized maintenance tasks in a simulated production line, though the project stopped before commercialization due to data integration challenges. Third, conceptual frameworks are already under discussion in many organizations (R1, R4, R5, R9), and digital twins are seen as one of the ideal tools for the ‘smart factory’ of the future, but a concrete implementation roadmap is lacking. Even companies that have already adopted the technology face persistent barriers: real-time data granularity limits the accuracy of dynamic models (R7), while organizational resistance slows the transition from manually driven diagnostics to AI-enhanced workflows (R6). As R6 summarized, “We’ve built the scaffolding for smart digital twins ... now we need to populate it with intelligence.”

4.2.2. Barriers and challenges

In general, there are four kinds of challenges and barriers: organizational barrier, skill gaps, data limitation, and digital twins’ life-cycle management.

Firstly, all respondents (R1–R9) identified organizational rather than technical barriers as their primary challenge. Cross-departmental collaboration emerged as a key issue, with R6 noting conflicting priorities hinder unified digital twin adoption, while R3 highlighted resistance to shifting decision-making power in traditional teams. R8 expected to have cross-functional teams for applying digital twins. Adoption level remains cautious overall. R4 observed most organizations settle for basic digital shadows, investing only when productivity gains are clear. R5 and R7 noted process inertia keeps AI-enhanced digital twins in pilot phases, despite available expertise. R9 pointed to unclear strategic goals for predictive maintenance applications. Trust barriers were particularly emphasized. One automotive expert (R3) explained their near real-time system deliberately avoids autonomous control due to security concerns: “We mirror PLC data...but can’t control machines via dashboards”. R3 added, “The real battle isn’t technology, but convincing 20,000 employees to trust AI decisions”.

Secondly, skill gaps limit progress, particularly in SMEs (R9), where few professionals possess hybrid expertise in domain-specific maintenance practices and AI techniques. There is a knowledge disconnect between engineering teams and field operators, with a shortage of people with a combination of knowledge of maintenance, digital twins, and algorithms (R2), and the development of a traditionally experienced workforce requires a systematic reshaping of the skills framework, but the cost and acceptance of large-scale training is preventing the technology from becoming routine (R5, R6).

Third, data limitation is one of the most serious technical challenges. The lack of data standardization contributes to the difficulty of creating digital twins (R2, R4). One manufacturing leader (R4) noted that high-fidelity digital twins require not only the volume of data, but also contextual metadata such as environmental conditions of manufacturing or maintenance history of machines, which are rarely captured by the system, and even when such data exists, it is of low quality and difficult to use to support the creation of digital twins. R7 mentioned that building robust, accurate digital twins requires a continuous, high-quality stream of data, but manufacturing environments typically deliver data in batches rather than in real time. “Even if the technology is in place, scaling it to thousands of machines exacerbates issues with storage, connectivity and modeling complexity”. R9 mentioned that his customers typically do not track or store relevant data with the frequency required for digital twins.

Fourth, digital twins’ life-cycle management challenges are underestimated: digital twins demand continuous updates as physical assets evolve, yet many companies lack protocols for model recalibration. A consultant (R4) warned that without such processes, “even the most advanced twin will degrade into a digital artifact of historical interest”. These challenges are worse in brownfield environments, where modifying old machines creates compatibility issues and technical debt (R5, R8).

4.2.3. Expectations and outlook

Interviewees identified three priorities for advancing AI-enhanced digital twins, focusing on foundational enablers, usability, and automation. First, standardization is essential to unlock scalability and integration. Participants (R1, R6) emphasized the need for unified data schemas and interoperability frameworks to connect models across diverse systems. Without such standards, organizations risk remaining isolated in “islands of automation”, as a simulation expert (R1) warned, making it difficult to aggregate data from various equipment providers. Second, human-centric design is critical for ensuring usability and adoption. R7 and R8 highlighted the importance of intuitive interfaces and immersive technologies like AR/VR to help technicians interpret AI outputs effectively. For instance, R7 posed questions about how best to present digital twin insights: “Should it be on the app? ... on the VR glasses? ... on something else?” Such considerations show the need for aligning technical solutions with real-world workflows. Third, participants expressed high expectations for generative AI in automating repetitive tasks and enhancing efficiency. R2, R8 noted its potential to accelerate processes such as code generation, synthetic data creation, and model updates, reducing reliance on specialized programmers. Automated code generation and natural-language interfaces (R2, R7) will empower domain experts to develop and customize digital twin models without requiring deep programming expertise. However, ethical and regulatory risks remain a significant concern. Under policies like the EU AI Act [146], governance protocols must ensure transparency and accountability, especially in safety-critical maintenance contexts, as an automotive practitioner (R3) stressed.

Participants also shared their outlook of AI-enhanced digital twins, anticipating a shift from pilots to industrial norms driven by two major trends. First, greenfield facilities are expected to adopt these technologies by default. Leveraging modular architectures and 5G-enabled IoT networks, such facilities will support real-time analytics from the outset (R5, R8). For example, an automotive practitioner (R5) expected that new battery factories will integrate digital twins during the design phase, unifying maintenance, logistics, and production planning into a cohesive model. Second, human-AI collaboration will transform how work is done. Technicians will shift into more strategic roles, supervising AI systems that support digital models, while they focus on higher-level decision-making and complex problem-solving (R6, R8). However, participants cautioned that only technological progress is insufficient. Workforce up-skilling and cultural adaptation are equally critical to ensure trust in AI outputs and avoid “automation paralysis”. R9 emphasized that employees must be equipped to interpret AI insights effectively.

The interviews collectively highlight AI-enhanced digital twins as a transformative yet underutilized technology. While advancements like generative AI and real-time IoT are expanding capabilities, systemic barriers, such as fragmented data governance and workforce readiness, continue to hinder adoption. Interviewees emphasized the importance of phased implementation, starting with robust data management and targeted workforce training to build foundational readiness. Additionally, participants stressed the urgency of developing interoperability standards and ethical frameworks to ensure responsible deployment and avoid fragmentation. By addressing these challenges, industries expect to move from isolated pilots towards fully integrated, smart maintenance ecosystems.

4.3. Gaps between academia and industry

There are several gaps between academic research and industry implementation and expectation about the current digital twins applications in maintenance, and the comparison based on three dimensions are shown at Table 3.

Table 3
Gap dimensions for academia research and industry implementation and expectation.

Dimensions	Academia	Industry
Scale	Small-scale experiments (1–10 machines); Controlled environment simulations; Limited physical constraints.	Large-scale deployments (100–1000+machines); Heterogeneous equipment fleets; Real-world dynamic environments.
Data	Clean, labeled benchmark datasets; Synthetic/idealized data generation; Limited temporal scope (months).	Noisy, incomplete real-time data streams; Proprietary data with access restrictions; Long-term operational data (yr).
Model	Accuracy metrics and performance; Minimal deployment constraints; Simulation dependent model validation approach.	Lightweight, explainable models; Trade-off between accuracy and computational cost; Robustness; Closed-loop model validation of physical entities.

4.3.1. Scale gaps

First, regarding research scale, academic studies mainly focus on controlled, small-scale experiments involving individual machines or laboratory setups [63,67,101,104]. While this approach ensures theoretical rigor, it often cannot accommodate the complexity of industrial deployments. Typically, these studies validate algorithms and digital twins on limited machines under idealized conditions, which limits their practical applicability [63,67,104]. Some researchers [82] have developed methods requiring minimal historical data, which are more viable for small to medium-sized factories, where predictive maintenance implementation in such environment is still a challenge.

In contrast, industrial applications require managing hundreds to thousands of interconnected machines within dynamic factory environments. For instance, practitioners from automotive manufacturing (R6, R7) emphasized that large-scale production lines may involve thousands robots, automated guided vehicles, and processing units operating simultaneously, with interdependencies that challenge traditional maintenance strategies. Newly built facilities, like a Nordic manufacturer’s battery factory (R5), face scalability challenges when deploying system-level digital twins across 1,000 machines. At scale, unanticipated interactions arise between equipment and environmental factors, such as temperature fluctuations and material flow bottlenecks.

The gap between academic prototypes and industrial expectations is further exacerbated by the need for different scale digital twins. While half of selected academic research prioritizes component or equipment level models, industry practitioners (R3, R7) stressed that system level twins which integrate production, logistics, and maintenance planning are critical for holistic optimization. However, simulating entire factories introduces “combinatorial complexity” (R6), where variables like equipment health, operator availability, and real-time order changes create a “sim-to-real credibility gap” (R2). For example, R7 noted that validating a predictive maintenance algorithm on a single robot in a lab provides little insight into its robustness across 4,000 robots with varying wear profiles and maintenance histories. This misalignment highlights the need for academia to adopt scalability-aware evaluation frameworks, such as hybrid testbeds that combine academic algorithms with industrial data pipelines, to bridge the “model-to-machine mismatch” (R4).

4.3.2. Data gaps

Significant differences exist in how academia and industry approach data for AI-enhanced digital twins in maintenance. Some selected studies [64,65,67,70,82,86,89] often uses clean, structured datasets collected from controlled lab environments or simulations, while in some academic research increasingly incorporates sensor-collected data from industrial equipment, for example, vibration signals [48,58,85,100, 104] or temperatures [59,60,73,74,95,100] to create digital twins. These datasets are often limited in scope and diversity compared to industrial deployments. For example, a university lab might train models on days or months of sensor data from a single production line or a small fleet of identical machines with a setup that ensures controlled

conditions but fails to capture the data diversity of industrial environments. In contrast, industries generate data streams from thousands of sensors across diverse, aging equipment with varying operating conditions. Practitioners emphasized challenges such as sensor noise (e.g., misaligned readings due to electromagnetic interference), missing data from disconnected legacy systems, and the difficulty of labeling years of historical maintenance logs (R5).

This gap is particularly evident in how “what-if” scenarios are validated. Academia often uses hybrid approaches that combining sensor data from physical prototypes with discrete event simulation to generate synthetic failure scenarios. For example, simulating a robotic arm’s motor degradation based on lab-collected vibration data [64]. However, industry experts (R6, R7) noted that such hybrid models struggle to replicate the emergent behaviors of large-scale systems.

Another critical difference is data duration and accessibility. Academic studies, even those using real sensors, typically analyze short-term datasets which are months, for example, study [67,75] used data from months scope, while industrial maintenance expects models trained on multi-year operational histories to capture rare failure modes or seasonal degradation patterns. For example, a Nordic manufacturer’s maintenance team (R7) noted that a bearing failure might occur once every 5–7 years in heavy machinery, making it impossible to detect with academic datasets that rarely exceed 12 months except some popular benchmarking dataset, for example, CMAPSS from NASA [147]. Additionally, confidential data restrictions limit collaboration: companies (R3, R5, R6) cannot share decade-long equipment logs with researchers due to intellectual property concerns, making academia lack of validation opportunities for long-term predictive models.

4.3.3. Model gaps

In academia, AI models are designed to optimize theoretical accuracy metrics and predictive performance, often tested in controlled environments. Researchers prioritize predictive capabilities and algorithmic complexity, aiming to reach and advance model’s optimal performance. For example, studies [59,62,66,69,77,96,106] emphasized the model accuracy and performance in their studies. Conversely, industry practitioners emphasize the practical utility of AI models, favoring lightweight, explainable, and actionable solutions that can seamlessly integrate into existing workflows (R2, R3, R6, R7).

In industrial settings, robustness and reliability under real-world conditions take precedence over theoretical performance metrics, as practical applications require models that can handle noisy data and dynamic operating scenarios. Several studies have conducted robustness-related experiments to strengthen industry confidence in these solutions [59,61,66,69,94]. Additionally, explainability is crucial in industry contexts to ensure trust and accessibility for diverse stakeholders, including operators and decision-makers, as R6 indicates: “if we have an alarm and AI says we have this problem ... the new guy goes forward and don’t know anything about it and asks ... it’s explained for him ... simple and effective ... easy, explainable AI.”

For model validation, industry expects to validate the AI and digital twin model through physical entities in closed loop (R3, R5),

while academia tends to validate through simulation dependent approach. Some studies attempt to bridge this gap by combining simulations with physical data. For example, Karkaria et al. [93] proposed a simulation-based verification framework that integrates simulation parameters while expanding the coverage of real-world working conditions, thereby increasing confidence in model validation.

5. Discussion

In the following sub-sections, we discuss the results of our study about AI categories and empowered capabilities for digital twins, gaps between academia studies and industry applications, and actionable recommendations for industry to better use academic research for their daily maintenance practices.

5.1. AI categorization and empowered capabilities for digital twins

In AI categorization to enhance digital twins, supervised learning remains the predominant approach for predictive maintenance in digital twin applications, whereas unsupervised learning is comparatively uncommon. The core objective of predictive maintenance, forecasting equipment failures or RUL, typically requires explicit labels such as failure state, failure type, or time to failure, and direct mappings between sensor data and target variables, tasks well suited to supervised methods. Historical fault records and maintenance logs in industrial settings can be linked to sensor measurements (temperature, vibration, pressure, etc.), forming labeled datasets that enable supervised learning algorithms such as random forest and XGBoost to achieve high predictive accuracy.

By contrast, unsupervised learning relies on unlabeled data to uncover latent patterns or structures, which cannot directly leverage the explicit fault labels required for predictive maintenance. Although it is effective for anomaly detection and clustering, unsupervised learning outputs typically need additional interpretation or post-processing before informing maintenance decisions. Furthermore, the high dimensionality, non-linearity, and noisiness of industrial data often exacerbate the limitations of unsupervised methods. Consequently, some researchers have adopted weakly supervised [87] or semi-supervised [102] techniques, combining the benefits of unlabeled data exploration with limited labeled data to enhance predictive maintenance tasks.

The reviewed studies reveal a predominant focus on enhancing predictive functionalities of digital twins, with limited attention directed toward their foundational creation. In data-driven approaches, synthetic data generation techniques, such as GANs and Variational Auto-Encoders (VAEs) are used to address data scarcity challenges by simulating physical system behaviors. For instance, GANs have been deployed to synthesize fatigue damage datasets for steel components, enabling the construction of high-fidelity digital twins when empirical data is unavailable [85]. In parallel, model-driven methodologies leverage machine learning to refine physical models. Techniques like model order reduction (MOR) simplify complex system representations into lightweight digital twins while preserving accuracy. A notable implementation employs MOR to derive real-time digital twins from intricate finite element models, ensuring computational efficiency without sacrificing predictive validity [73]. These hybrid strategies emphasize the role of algorithmic calibration in aligning digital twins with real-world dynamics.

Beyond static modeling, advanced algorithms enable dynamic adaptation of digital twins to evolving system states. Temporal Fusion Transformers (TFTs) exemplify this capability, integrating real-time sensor data to iteratively update digital twin parameters and keep them in sync with physical counterparts [93]. Structural automation further enhances scalability, as demonstrated by neural network-driven frameworks that convert manual maintenance logs into adaptive digital twin architectures. For example, RNNs transform maintenance timestamps and failure frequencies into iterative optimization processes,

refining equipment life-cycle management [90]. Self-learning mechanisms, such as NARX models, detect tool wear anomalies through vibration analysis, triggering autonomous model recalibration [48]. These innovations highlight a paradigm shift toward self-optimizing digital twins, where algorithmic feedback loops, grounded in Bayesian optimization or reinforcement learning are continuously refine system representations to align with operational realities [76,84].

5.2. Frameworks of AI enhanced digital twins for maintenance

Out of the 51 studies reviewed, 19 studies proposed their own digital twin frameworks. These frameworks vary in their design and emphasis, particularly in terms of architecture, AI integration, and implementation approach. One common category is centralized digital twins with cloud-based AI, where a single, cloud-hosted digital twin manages data from all assets and runs AI models centrally. For instance, Zhang et al. [88] proposed a global digital twin for power equipment with machine learning models deployed on the cloud, while Guc and Chen [80] utilized a cloud environment to aggregate data and train advanced AI models for fault classification. Centralized frameworks are advantageous due to their simplicity in orchestration, relying on a single platform, and scalability, as cloud resources can handle large datasets efficiently. However, they may encounter challenges in latency-sensitive applications due to their dependence on centralized processing.

In contrast, other frameworks adopt distributed or hybrid edge-cloud architectures, where real-time analytics are performed on edge nodes located near physical machines, while computationally intensive tasks such as AI model training or simulations are handled in the cloud. For example, Bellavista et al. [54] deployed hybrid digital twins at local edge nodes for immediate anomaly detection, while Rossini et al. [56] utilized distributed simulation environments and AI modules positioned across different network locations. These distributed frameworks excel in reducing latency and minimizing cloud bandwidth usage by processing or aggregating data locally. However, they require more complex architectures with robust orchestration between edge and cloud components to ensure seamless functionality. Regardless of whether a framework is centralized or distributed, the Industrial IoT (IIoT) layer remains fundamental to all digital twin implementations. This layer serves as the bridge between physical systems and their virtual counterparts by collecting real-time sensor data and enabling communication between physical and digital entities.

Based on summarization and analysis of these individual frameworks, we proposed a generalized five-layer architecture framework to consolidate common functionalities across studies at Fig. 7.

This architecture includes the following layers: The physical layer, which captures system-level and equipment-level data directly from machines using sensors. It will provide sensor data stream and equipment status signals to data collection and transmission layer, which incorporates sensors, communication protocols, and computing services (e.g., edge or cloud computing) for securely transmitting data. Then the preprocessed sensor data and eventlogs with timestamps will be sent to digital twin layer, which maintains synchronized virtual models using simulation software or IIoT platforms. Digital twin layer will provide simulation results and state information to AI analytics layer and visualization of asset condition to maintenance service layer. AI analytics layer processes data from the digital twin layer to enhance predictive capabilities using machine learning and other AI techniques. Finally, the maintenance service layer, which provides actionable services such as fault diagnosis and anomaly detection, and it will send back the maintenance actions to physical layer, for example, component replacement.

While some frameworks from selected studies incorporate all five layers comprehensively, others focus on specific aspects depending on their objectives. If a digital twin system for maintenance needs to realize full functionalities, for example, seamless integration, visualization

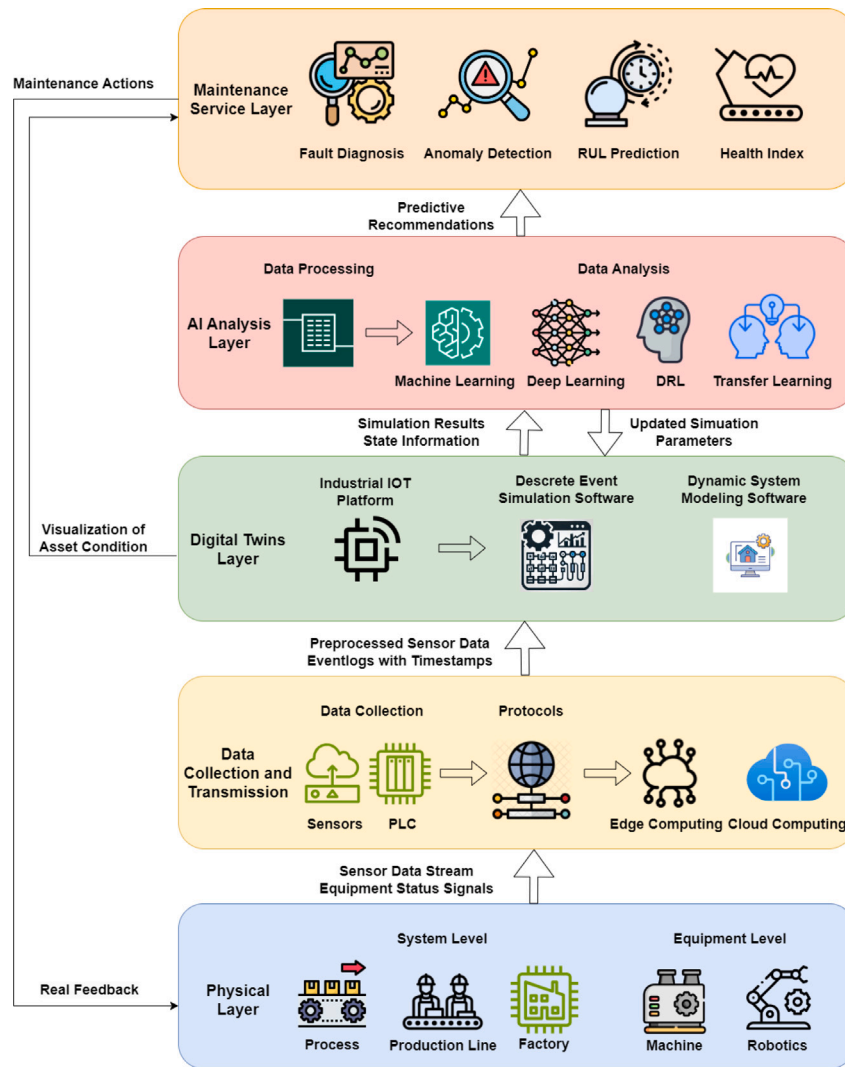


Fig. 7. Framework of AI enhanced digital twins for maintenance practices.

and in-time prediction, the digital twin framework will cover all 5 layers. Nevertheless, the digital twin layer and its integration with IIoT remain necessary to all implementations.

The overall purpose of these AI-enhanced digital twin frameworks is to improve the efficiency, reliability, and cost-effectiveness of maintenance processes by combining real-time data acquisition, advanced analytics, and virtual modeling. By enabling predictive and prescriptive maintenance through techniques such as machine learning, deep learning, and reinforcement learning, these frameworks forecast equipment failures, optimize maintenance schedules, and reduce downtime. For instance, hybrid edge-cloud architectures like those in IoTwins [54,55] and REPLICA [56,57] provide real-time fault detection alongside advanced AI-driven insights. In industries such as aviation, automotive, manufacturing, and energy, digital twins simulate physical systems under various conditions to ensure proactive responses to potential issues. By integrating IoT sensors with AI algorithms, these frameworks enhance decision-making capabilities by dynamically adapting to operational changes while minimizing disruptions. Furthermore, they contribute to sustainability by extending equipment lifespans through optimized resource allocation and reduced waste.

5.3. Gaps discussion & actionable recommendations

Beyond the three primary gaps (scale, data and model) between academia and industry, several additional gaps also need to be attention. First, while industry practitioners emphasize organizational and management challenges, this perspective remains notably absent in academic studies. This shows that academic research cannot be considered only from technical point of view, but also from a managerial and coordinating point of view. Second, workforce skill gaps emerge as a critical constraint for technology adoption in industry, yet academic research overlooks this practical consideration. Notably, none of the selected studies address labor readiness when implementing AI-enhanced digital twins for maintenance. Technicians and maintenance staff who already understand maintenance but lack expertise in digital twins and AI will need proper training to bridge this gap. Finally, a clear divergence exists in life cycle perspectives. Industry prioritizes comprehensive digital twin life cycle management, whereas academic studies, despite their long-term view, rarely explicitly incorporate this dimension.

Bridging the gap between academic research and industry expectations requires creating scalable frameworks that fit the reality of industrial operations, while our summarized framework in Fig. 7 is one

feasible example. To address high scale dimension challenges, industry should adopt hierarchical and modular digital twin architectures that decompose complex manufacturing systems into sub-systems while containing all five layers components from the framework. A progressive application strategy can begin with pilot projects on key equipment or production lines, and then be expanded to the entire factory, achieving expansion from a low scale dimension to a high scale dimension. Edge-cloud hybrid computing architectures can distribute computational loads, with edge devices handling real-time local processing while cloud infrastructure manages complex analytics and machine learning model training. Moreover, industrial cloud platforms equipped with adaptive knowledge transfer capabilities can enhance system resilience at high scales while simultaneously reducing implementation barriers and improving industrial acceptance through intelligent optimization algorithms [143,144].

Data integration complexities can be mitigated through standardized data exchange protocols and industrial communication standards (such as OPC UA and MQTT). Selection of proper digital twins platform optimized for scalability of AI is also important. While addressing insufficient workforce readiness, it requires comprehensive upskilling programs that combine theoretical knowledge with hands-on experience using digital twins supported maintenance scenarios. Virtual and augmented reality (VR/AR/XR) training platforms can provide immersive learning environments where technicians can practice complex maintenance procedures on digital twin simulations without risking equipment damage or production disruption.

Researchers can collaborate with industry to develop hybrid testbeds that combine lab-developed models with real industrial data. These testbeds can simulate large-scale manufacturing environments by incorporating different machine configurations, relations, and dynamic operating conditions. For example, in collaboration with manufacturers, academic prototypes can be deployed in industrial settings, allowing researchers to validate models on interconnected machines with realistic constraints. By addressing the complex interactions between equipment and environmental factors in large-scale systems, these system-level twins can provide practical information to optimize maintenance strategies while ensuring the robustness of industrial operations.

Academia and industry can establish long-term data-sharing frameworks and improve methods for handling noisy or incomplete datasets. Secure mechanisms like federated learning or blockchain-based platforms can facilitate access to anonymized multi-year datasets without compromising confidentiality [148]. Such datasets are important for training AI models capable of capturing rare failure modes and long-term degradation patterns often missing in short-term studies. Academia should also prioritize advanced preprocessing techniques to address sensor noise and missing data prevalent in industrial environments. For example, generative AI could augment historical maintenance logs to fill data gaps and improve model robustness [149]. Finally, adopting hybrid validation approaches that combine real-world sensor data with synthetic failure scenarios would ensure predictive models are tested against both controlled conditions and emergent behaviors in complex systems.

5.4. Research limitation

One limitation of our research is the geographical scope of the interviews, which were conducted exclusively with industrial practitioners in Sweden. Although Sweden has a strong manufacturing tradition, future studies could expand to other countries or regions where industrial practices, technological adoption, and maintenance strategies may differ due to variations in regulatory frameworks and industry conditions. While clear signals of saturation were observed from nine interviews, we acknowledge that similar qualitative studies often employ slightly larger sample sizes to establish robust saturation [150]. Future research should therefore focus on extending the

interview scope to include multiple countries, diverse industries, and an expanded participant pool.

Another limitation lies in the potential biases introduced by relying on interviews. While interviews provide rich insights into industry practices and challenges, they are inherently subjective and may reflect individual experiences rather than broader trends. In the future, quantitative validation can increase the robustness of conclusions regarding long-term trends and the scalability of proposed solutions.

6. Conclusion

This study systematically investigates the evolving role of AI-enhanced digital twins in supporting maintenance decision-making and addresses the gaps between academic research and industrial practices by combining a SLR with interviews conducted with industrial practitioners. First, it identifies the application domains, maintenance objectives, common digital twin platforms, and categories of AI models that empower digital twins. From an industrial perspective, the study explores the current status of AI-enhanced digital twin applications, the barriers and challenges hindering their adoption, and the expectations of industrial stakeholders. Finally, it highlights three key dimensions: scale, data and model gaps between academic research and industrial implementation and expectation, offering actionable recommendations and proposing a general framework derived from selected academic studies for implementing AI-enhanced digital twins in maintenance.

This study contributes to advancing maintenance practices by facilitating the transition from predictive to prescriptive maintenance. Ultimately, it aims to accelerate the transformation of AI-enhanced digital twins from theoretical concepts into practical applications that drive operational efficiency and foster sustainable growth across industries.

CRediT authorship contribution statement

Siyuan Chen: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ebru Turanoglu Bekar:** Writing – review & editing, Supervision. **Jon Bokrantz:** Writing – review & editing, Supervision. **Anders Skoogh:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

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Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT 4o in order to proofread and enhance readability. After using this tool, the authors reviewed and edited the content and take full responsibility.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table B.4

Classification of research work of AI-enhanced digital twin for maintenance.

Title & Reference	Year	Industry	Physical system	AI techniques	Maintenance issues
Digital twin for machining tool condition prediction [58]	2019	Industrial IoT & Machinery	Milling Machine	Deep Stacked GRU (DSGRU), Bi-directional GRU	DSGRU supports a five-dimensional digital twin for milling tool-wear prediction using vibration data
A digital-twin-assisted fault diagnosis using deep transfer learning [72]	2019	Automotive	Car Body Production Line	Deep transfer learning (Stacked Sparse Autoencoders + transfer)	Two-phase DT-assisted fault diagnosis (DFDD) in development and maintenance phases
Edge Cloud as an Enabler for Distributed AI in Industrial IoT Applications: the Experience of the IoTwins Project [54]	2019	Smart Factories & Production Lines	Manufacturing Process	Distributed AI (decentralized edge/cloud learning)	Predictive maintenance to minimize downtime; real-time process/fault predictions
Digital twin of subsea pipelines: conceptual design integrating IoT, machine learning and data analytics [69]	2019	Energy	Subsea Pipeline	LSTM	Fatigue life estimation for subsea pipelines
Cloud-based digital twin for robot integration in intelligent manufacturing systems [100]	2020	Robotics	SCARA Robot	Deep neural networks, classification methods, clustering approaches	Anomaly detection and tailored maintenance programs
REPLICA: A solution for next generation IoT and digital twin based fault diagnosis and predictive maintenance [56]	2020	Smart Factories & Production Lines	Large Woodworking Production Line	AI modules (Acumos AI)	Fault diagnosis and predictive maintenance
The use of the “digital twin” concept for proactive diagnosis of technological packaging systems [86]	2020	Smart Factories & Production Lines	Packaging Systems	Statistical Models	Mathematical model to determine machine state and predict necessary preventive maintenance actions
An architecture based on digital twins for smart power distribution system [88]	2020	Energy	Power Distribution System	Cloud Computing	Transformer performance anomaly detection and predictive maintenance
Real-world anomaly detection by using digital twin systems and weakly supervised learning [87]	2021	Smart Factories & Production Lines	Machinery	Clustering, Weakly Supervised Learning, Siamese Autoencoders (SAE)	Facility monitoring anomaly detection (digital twin data + minimal labeled anomalies)
Digital twin-driven online anomaly detection for an automation system based on edge intelligence [76]	2021	Industrial IoT & Machinery	Absorption Chiller	Supervised ML-based anomaly detection; Cloud/Edge computing	Digital twin framework for anomaly detection, leveraging edge computing for real-time insights
Digital twin for offshore pipeline corrosion monitoring: A deep learning approach [66]	2021	Energy	Pipelines	CNN	Pipeline corrosion detection and classification
A prescriptive maintenance system for intelligent production planning and control in a smart cyber–physical production line [65]	2021	Smart Factories & Production Lines	Production Line	ML-based predictive analytics	Integrates production scheduling with maintenance in a make-to-order environment
Toward digital twin for cyber physical production systems maintenance: observation framework based on AI techniques [70]	2021	Smart Factories & Production Lines	Learning Factory	Logistic Regression, ANN, SVM	Predictive maintenance in a learning factory; uses distribution error analysis
Holistic concept towards a reference architecture model for predictive maintenance [68]	2021	Industrial IoT & Machinery	Axis Test Rig	SVM, Autoencoder, LSTM	Predict spindle/axis damage for proactive scheduling
Digital twin-driven decision support system for opportunistic preventive maintenance scheduling in manufacturing [63]	2021	Smart Factories & Production Lines	Shop Floor	No specific named algorithm	Opportunistic preventive maintenance scheduling to minimize throughput penalties
Increase food production efficiency using the executable digital twin (xDT) [73]	2021	Food Industry	Ketchup Production	Reinforcement learning for control	Improved temperature monitoring/control; RL-based optimization to avoid jams
AI environment for predictive maintenance in a manufacturing scenario [57]	2021	Smart Factories & Production Lines	Kitchen Production Line	Multiple integrated AI modules	Fault detection, refurbishment planning, and parameter optimization
Intelligent fault diagnosis of machinery using digital twin-assisted deep transfer learning [77]	2021	Industrial IoT & Machinery	Triplex Pump	Deep transfer learning, Novel Sparse de-noising Autoencoder	Machinery fault diagnosis for triplex pumps
Predictive maintenance framework for production environments using digital twin [97]	2021	Automotive	Hydraulic Accumulator	Gaussian Naive Bayes (GNB), Adaptive Random Forest (ARF), LightGBM (LGBM)	Identifies hydraulic accumulator failure; predicts turbofan engine degradation
Explorative hybrid digital twin framework for predictive maintenance in steel industry [74]	2022	Steel Industry	Steel Manufacturing Line	XGBoost	Detects false faults in hot rolling mill cages
Digital twin and machine learning for decision support in thermal power plant with combustion engines [71]	2022	Energy	Power Plants	ANN, Gradient Boosting, Wavelet Transform	Reduced corrective downtime; fault/failure classification
Machine-learning-driven digital twin for lifecycle management of complex equipment [59]	2022	Railway Industry	Diesel Locomotive	LASSO, SVR, XGBoost	Alarms for abnormal axle temperature; lifecycle health management
Data fusion for predictive maintenance of industrial assets using digital twin [78]	2022	Industrial IoT & Machinery	Metadata and Telemetry Data	SVM, Random Forest, Logistic Regression, KNN, Naive Bayes, Decision Tree	Fault diagnosis via fused telemetry data

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Table B.4 (continued).

Title & Reference	Year	Industry	Physical system	AI techniques	Maintenance issues
Feature investigation with digital twin for predictive maintenance following a machine learning approach [75]	2022	Steel Industry	Steel Production Line	XGBoost	Steel production anomaly detection
lotwins: Toward implementation of distributed digital twins in Industry 4.0 settings [55]	2022	Smart Factories & Production Lines	Machining Processes	Hybrid/Distributed ML (edge + cloud)	Time-series error detection and predictive insights for manufacturing operations
Predictive maintenance based on machine learning model [79]	2022	Industrial IoT & Machinery	Ball Bearing	RNN, CNN, Temporal CNN (TCN)	RUL prediction for ball bearings
Smart predictive maintenance enabled by digital twins and smart big data: A new framework [80]	2022	Industrial IoT & Machinery	Mechatronic Test Bed	MATLAB Diagnostic Feature Designer, Decision Tree	RFIM fault classification via big data analytics
Circuit Design for Predictive Maintenance [81]	2022	Industrial IoT & Machinery	Circuit	Circuit-level AI solutions	System-level improvements (nonlinearities, control-loop constraints)
Digital twins for predictive maintenance [82]	2023	Industrial IoT & Machinery	Racing Bicycles Factory	Statistical RUL models	Identifies fault types, predicts degradation paths, and estimates RUL
Simulation modeling and temperature over-advance perception of mine hoist system based on digital twin technology [60]	2023	Mining	Hoist Motor	RNN, LSTM	Fault diagnosis framework using temperature prediction
Digital Twin in Oil and Gas Equipment for Predictive Maintenance [89]	2023	Energy	Drilling System	Machine learning	Fault detection and RUL estimation (drilling equipment)
Digital Twin Approach to Support Preventative Maintenance in a Robotic Application [64]	2023	Robotics	Robotic Arm	Machine Learning	Monitoring robotic processes for preventive maintenance
A method for Generating a Digital Twin Structure for a System for Organizing Preventive Maintenance in the Electricity Sector [90]	2023	Energy	Electric Nodes	Neural networks	Scheduled preventive maintenance for selected power devices
A Digital Failure Twin Model For PHM: From Concepts To Maturity Levels [83]	2023	Industrial IoT & Machinery	Conceptual System	Various advanced ML	PHM framework to predict RUL and plan preventive maintenance
Artificial intelligence enabled digital twin For predictive maintenance in industrial automation system: a novel framework and case study [101]	2023	Lab	DC Motor	NARX neural network	Anomaly detection, demonstration of digital twin-based approach
Vibration signal-based fault prediction using few-shot learning in the digital twin ecosystem [104]	2023	Lab	Vibration Sensor	Siamese Neural Network, Few-shot Learning, SVM	Converts vibration signals to images for fault identification and classification
Robotic cell reliability optimization based on digital twin and predictive maintenance [106]	2023	Robotics	Robotic System	SVM classification (medium Gaussian SVM)	RUL prediction to improve reliability of robotic cells
Dynamic Maintenance Management Approach Based on Real Time Monitoring and Artificial Intelligence Using Digital Twins [95]	2023	Railway Industry	Train Axle Bearings	Neural network	Train maintenance scheduling, bearing temperature monitoring (RUL), automatic vision-based inspection
Railway infrastructure maintenance efficiency improvement using deep reinforcement learning integrated with digital twin based on track geometry and component defects [96]	2023	Railway Industry	Railway Track	Advantage Actor Critic (A2C), Deep RL	Railway maintenance optimization (reducing defects/costs)
Prescriptive maintenance for complex products with digital twin considering production planning and resource constraints [61]	2023	Aviation	Aero-engine	Stochastic programming (risk-based), e.g., DPformer	Aero-engine prescriptive, RUL-based maintenance scheduling
Unsupervised anomaly detection in pressurized water reactor digital twins using autoencoder neural networks [91]	2023	Energy	Nuclear Plant	Autoencoder, unsupervised learning	Anomaly detection for reactor scram intervention
Predictive maintenance of standalone steel industrial components powered by a dynamic reliability digital twin model with artificial intelligence [85]	2024	Steel Industry	Steel Components	Boosted Decision Tree Regression, Decision Forest Regression, Linear Regression, Poisson Regression, Neural Network	Calculation of steel component RUL (fatigue and reliability modeling)
Experience from implementing digital twins for maintenance in industrial processes [13]	2024	Steel Industry	Machinery Tools	Regression analysis	Maintenance needs detection, tool degradation analysis, and RUL estimation
Immune system inspired smart maintenance framework: tool wear monitoring use case [84]	2024	Industrial IoT & Machinery	CNC Milling Machine	Clustering algorithm	Tool wear monitoring with an immuno-like maintenance framework
Towards a Distributed Digital Twin Framework for Predictive Maintenance in Industrial Internet of Things (IIoT) [92]	2024	Energy	Wind Turbine	LSTM, XGBoost, RNN, Multiple Linear Regression	Wind turbine anomaly detection, condition monitoring for predictive maintenance
Enhancing Digital Twins With Deep Reinforcement Learning: A Use Case in Maintenance Prioritization [67]	2024	Lab	Lab-scale Drone Assembly Stations	Proximal Policy Optimization (PPO)	Maintenance prioritization using DRL integrated with digital twin

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Table B.4 (continued).

Title & Reference	Year	Industry	Physical system	AI techniques	Maintenance issues
A Digital Twin Framework Utilizing Machine Learning for Robust Predictive Maintenance: Enhancing Tire Health Monitoring [93]	2024	Automotive	Automotive Tires	Temporal Fusion Transformer (TFT)	Monitoring tire health and remaining casing potential (RCP)
Fault diagnosis method for oil-immersed transformers integrated digital twin model [62]	2024	Energy	Oil-immersed Transformers	INNOS, SPCA, Aquila Optimizer (AO), KELM	Online transformer fault diagnosis
A semi-supervised method for digital twin-enabled predictive maintenance in the building industry [102]	2024	Building Industry	Building Facilities	Semi-supervised GAN	Fault prediction for critical facilities (HVAC, elevators, etc.)
Digital Twins for Automotive Predictive Maintenance [94]	2024	Automotive	Vehicle Components	Isolation Forest, LSTM, Random Forest, Gradient Boosting	Real-time monitoring and fault forecasting of engine/HVAC components
Digital twin-based anomaly detection for real-time tool condition monitoring in machining [48]	2024	Industrial IoT & Machinery	Machine Tool and Cutting Tool	Nonlinear AutoRegressive with eXogenous (NARX)	Real-time tool health monitoring

Appendix A. Full interview questions

- Interviewee’s background, industry, and role.
 - Could you briefly describe your background, role, and experience in your current industry?
 - What types of systems or equipment does your team manage for maintenance? Are there any specific industries or use cases you specialize in?
- Explore experience with digital twins.
 - Are digital twins currently part of your maintenance strategy? If so, can you share a specific example of how they have been applied?
 - At what level (equipment/system) do you feel digital twins are most effective for maintenance?
- Clarify the role of AI in maintenance.
 - How does your organization currently use AI in maintenance processes?
 - If AI tools are integrated, are they primarily used for prediction, decision support, or other functions? Are there digital twins involved in?
- Identify challenges (tailored to practical insights).
 - From your perspective, what are the main barriers (technical, organizational, or regulatory) to adopting digital twins and AI in maintenance?
 - Are there gaps in skills or resources that prevent the effective deployment of AI-enhanced digital twins?
- Future directions and desired advancements.
 - In your opinion, what are important improvements needed in digital twins to make them more applicable to maintenance?
 - How do you see AI enhanced digital twins evolving in your industry over the next 5–10 years?
 - What do you think about Generative AI in this field? Do you think it is a good tool for supporting digital twins in this case?
- Add flexibility for unanticipated insights.
 - Is there anything else you had like to share about your experience with AI or digital twins in maintenance that we have not covered?

Appendix B. Selected research works

See Table B.4.

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